



Business Context

AllLife Bank is a mid-sized, fast-growing US-based financial institution that offers a range of retail banking services, including savings and checking accounts, fixed deposits, and personal loans. The bank's business model is centered on building long-term customer relationships, expanding its retail footprint, and growing its loan portfolio to drive sustainable profitability through interest income.

It currently relies on a large base of liability customers (depositors) but faces a significant under-representation of asset customers (borrowers). To drive profitability through interest income, the bank must aggressively expand its loan portfolio by converting existing depositors into personal loan customers.

Last year's pilot campaign achieved a 9% conversion rate, validating the potential of this strategy. However, to optimize marketing spend and improve efficiency, the retail marketing department requires a more data-driven approach. Enhancing the success ratio of these campaigns is critical for sustainable growth and maximizing customer lifetime value.

Objective

The objective is to develop a predictive classification model that identifies patterns and key factors driving personal loan adoption among existing liability customers. By uncovering the demographic and behavioral drivers of loan conversion, the goal is to enable targeted segmentation and more precise marketing interventions that improve campaign conversion rates, optimize marketing spend, and enhance overall profitability through higher-quality loan portfolio growth.

Data Dictionary

- ID: Customer ID
- Age: Customer's age in completed years
- Experience: # years of professional experience
- Income: Annual income of the customer (in thousand dollars)
- ZIP Code: Home Address ZIP code.
- Family: The family size of the customer
- CCAvg: Average spending on credit cards per month (in thousand dollars)
- Education: Education Level. 1: Undergrad; 2: Graduate; 3: Advanced/Professional
- Mortgage: Value of house mortgage if any. (in thousand dollars)
- Personal_Loan: Did this customer accept the personal loan offered in the last campaign?
- Securities_Account: Does the customer have a securities account with the bank?
- CD_Account: Does the customer have a certificate of deposit (CD) account with the bank?
- Online: Do customers use Internet banking facilities?
- CreditCard: Does the customer use a credit card issued by any other Bank (excluding All Life Bank)?

Submission Guidelines

1. There are two ways to work on this project:

i. Full-code way: The full code way is to write the solution code from scratch and only submit a final Python notebook with all the insights and observations.

ii. Low-code way. The low-code way is to use an existing solution notebook template to build the solution and then submit a business presentation with insights and recommendations.

The primary purpose of providing these two options is to allow learners to opt for the approach that aligns with their learning aspirations and outcomes. The below table elaborates on these two options.

Submission type	Who should choose	What is the same across the two	What is different between the two	Final submission file [IMP]	Submission Format
Full-code	Learners who aspire to be in hands-on coding roles in the future focussed on building solution codes from scratch	Perform exploratory data analysis to identify insights and recommendations for the problem	Focus on code writing: 10 - 20% grading on the quality of the final code submitted	Solution Python notebook from the full-code template submitted in .html format	.html
Low-code	Learners who aspire to be in managerial roles in the future - focus on solution review, interpretation, recommendations, and communicating with business		Focus on business presentation: 10 - 20% grading on the quality of the final business presentation submitted	Business presentation in .pdf format with problem definition, insights, and recommendations	.pdf

Please follow the below steps to complete the assessment. Kindly note that if you submit a presentation, ONLY the presentation will be evaluated. Please make sure that all the sections mentioned in the rubric have been covered in your submission.

i. Full-code version

- Download the full-code version of the learner notebook from Olympus.
- Follow the instructions in that notebook to complete the project.
- Then download the Python notebook from Colab as a .ipynb file.
- Finally, convert and export the .ipynb file to HTML using this link:
- https://htmltopdf.herokuapp.com/ipynbviewer/#google_vignette

ii. Low-code version

- Download the low-code version of the learner notebook.
- Follow the instructions provided in the notebook to complete the project.
- Prepare a business presentation with insights and recommendations for the business problem.
- Submit only the presentation [format: .pdf]

2. Any assignment found copied/plagiarized with other submissions will not be graded and awarded zero marks.

3. Please ensure timely submission as any submission post-deadline will not be accepted for evaluation.

4. Submission will not be evaluated if

- it is submitted post-deadline, or
- if more than 1 file is submitted.

Best Practices for Full-code Submissions

- The final Python notebook should be well-documented, with inline comments explaining the functionality of code and markdown cells containing comments on the observations and insights.
- The notebook should be run from start to finish sequentially before submission.
- It is important to remove all warnings and errors before submission.
- The notebook should be submitted as an HTML file (.html) and NOT as a notebook file (.ipynb).
- Please refer to the FAQ page for common project-related queries.

Best Practices for Low-code Submissions

- The presentation should be made keeping in mind that the audience will be the Data Science lead of a company.
- The key points in the presentation should be the following:
 - Business Overview of the problem and solution approach
 - Key findings and insights that can drive business decisions
 - Business recommendations
 - Focus on explaining the key takeaways in an easy-to-understand manner.
 - The inclusion of the potential benefits of implementing the solution will give you the edge.
- Copying and pasting from the notebook is not a good idea, and it is better to avoid showing codes unless they are the focal point of your presentation.
- The presentation should be submitted as a PDF file (.pdf) and NOT as a .pptx file.
- Please refer to the FAQ page for common project-related queries.

Power Ahead!



Criteria

Exploratory Data Analysis

- Problem definition, questions to be answered
- Data background and contents
- Answer all the key questions asked in this section
- Univariate analysis
- Bivariate analysis
- Key meaningful observations for each of the plot

Comments

Areas Performed Well:

Clearly articulated the problem with well-defined objectives and insightful questions.

Demonstrated a deep understanding of the dataset, addressing potential biases and limitations.

Answered all the questions asked in this section

Conducted comprehensive univariate and bivariate analysis using diverse statistical and visualization techniques.

Effectively explained correlations, providing meaningful insights that strongly link EDA findings to the problem statement.

Points

10/10

Criteria

Data Preprocessing

- Anomalous Value Detection and Treatment (if needed)
- Missing Value Treatment (if needed)
- Outlier Detection and Treatment (if needed)
- Feature Engineering (if needed)
- Data Preparation for Modeling

*In case one or more of the above steps are not needed, please share the rationale for the same in your submission

Comments

Areas Performed Well:

Ensured data completeness by checking for missing values.

Properly split the data into train and test sets while maintaining a balanced distribution.

Performed outlier detection to enhance data quality and model robustness.

Handled experience data effectively, ensuring consistency.

Points

6/6

Criteria

Model Building

- Choose the metric of choice with proper rationale
- Build the decision tree model and comment on the model performance
- Visualize and comment on the decision rules of the model

Comments

Areas Performed Well:

Clearly defined the model evaluation criteria, ensuring alignment with the problem objectives.
Successfully built the model with insightful performance analysis.

Improvement:

Visualize decision rules making the model more interpretable.

Points

9/10

Criteria

Model Performance Improvement and Final Model Selection

- Try and improve the model performance by pruning (pre-pruning and post-pruning)
- Check the performance of the pruned models
- Compare the performance of all the models built
- Select the final model with proper rationale
- Visualize and comment on the decision rules and feature importance for the final model

Comments

Areas Performed Well:

Successfully applied pre-pruning and post-pruning techniques to enhance model performance.
Conducted a comprehensive model comparison, ensuring a data-driven final selection.
Effectively identified key decision rules and provided a clear feature importance analysis.

Points

20/20

Criteria

Actionable Insights & Recommendations

- Conclude with key takeaways for the business - what would your advice to the marketing team be regarding campaign management?

Comments

Areas Performed Well:

Clearly summarized key insights from the analysis in a way that is actionable for the marketing team.

Provided well-thought-out recommendations on how to execute the campaign based on data-driven findings.

Points

6/6

Criteria

Presentation/Notebook - Overall Quality

- Structure and flow
- Crispness
- Visual appeal
- All key insights and recommendations covered?

OR

- Structure and flow
- Well commented code
- All key insights and recommendations covered?

Comments

Areas Performed Well:

Well-organized structure and logical flow, making the analysis easy to follow.

Clear and concise content, effectively conveying key insights without unnecessary details.

Visually engaging notebook with well-selected plots and professional formatting.

Points

8/8